INCH-POUND

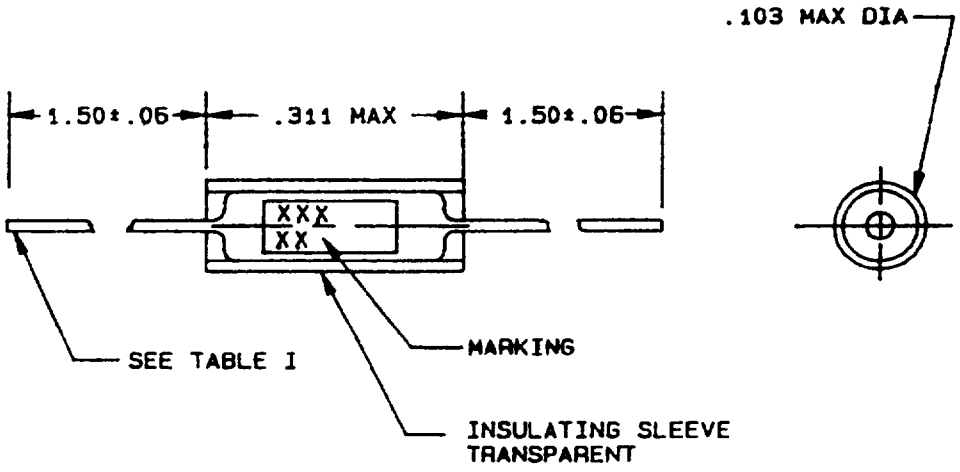
MIL-F-23419/11A(NA)
23 November 1992
SUPERSEDING
MIL-F-23419/11(NA)
22 December 1989

MILITARY SPECIFICATION SHEET

FUSE, CARTRIDGE, INSTRUMENT TYPE, STYLE FM11
(SUBMINIATURE - HIGH PERFORMANCE)

This specification is approved for use by National Aeronautics and Space Administration and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the fuses described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-F-23419



Inches	mm
06	1.5
103	2.62
311	7.90
1.50	38.10

NOTES:
1 Dimensions are in inches
2 Metric equivalents are given for general information only

FIGURE 1 Style FM11 fuse.

DISTRIBUTION STATEMENT A Approved for public release; distribution is unlimited

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REQUIREMENTS:

Parts supplied to this specification shall be qualified to MIL-F-23419/11, style FM11

Dimensions and configuration. See figure 1.

Applicable fuseholder. Grayhill type 2-42, miniature test clip or equivalent.

For 7, 10, and 15 ampere fuses use Grayhill push-post number 29YY2126-0 or equivalent.

Case material: Ceramic shall be grade L211 or higher grade in accordance with MIL-I-10.

Terminals:

Materials: End cap; 90/10, Bronze or CDA-10200 copper. Leads, CDA-10200 copper.

Finish: Leads gold plated. Gold plating shall conform to MIL-G-45204, type II, grade B, class 1. Nickel plating shall be in accordance with QQ-N-290 and shall be not less than .000030 inch (0.00076 mm) thick over brass and .0002 inch (0.010 mm) over copper

Strength: 10 pounds along terminal axis.

Current rating, nominal: See table I.

Characteristic: See table I.

Voltage rating: See table I.

Current carrying capacity: 100 percent at +25°C; 110 percent at -55°C; 80 percent at +125°C. The temperature of the case, body, or terminals shall, at no point rise more than +70°C above the ambient air temperature. The maximum temperature rise for 10 to 15 ampere fuses shall be +85°C.

Resistance: See table I.

Overload interrupt: Percentage of nominal rating, interrupt time -55°C through +125°C, 200 percent in 0 to 5 seconds; 300 percent in 0 to 0.1 seconds. Interrupt time for 15 ampere fuses shall be 10 seconds maximum at 200 percent of rated current and .3 second maximum at 300 percent of rated current.

Short circuit: 300 amperes at maximum voltage dc, 50 amperes at a power factor of 1.0 at maximum voltage ac.

Insulation resistance: Fuses shall be tested in accordance with MIL-F-23419, minimum insulation resistance, after overload interrupt shall be 0.5 megohm. Insulation resistance after short circuit tests shall be 10 kilohms.

Shock: In accordance with method I of MIL-F-23419.

Thermal shock: In accordance with MIL-F-23419, test condition B, method 107 of MIL-STD-202

Salt spray: Method 101, test condition A of MIL-STD-202.

Vibration (high frequency): Method 204, test condition D of MIL-STD-202. The applied current during test shall be 50 percent of rated current.

Random vibration: Method 214, test condition I, letter E, 15 minutes minimum per axis of MIL-STD-202.

Outgassing: Materials used in the finished fuse shall be tested in accordance with ASTM E595. Samples to be tested shall have been processed in the same manner as that used in production of the qualification lot. Materials when tested shall not release greater than 1.0 percent total mass loss, (TLM) and 0.1 percent collected volatile condensable material (CVCM). Data listed in NASA Reference Publication 1124 revised may be used in lieu of the actual test data for applicable materials.

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Solderability: Fuses shall be tested in accordance with MIL-F-23419.

Resistance to soldering heat: Fuses shall be tested in accordance with MIL-F-23419.

TABLE I. Type designation identification.

Style	Characteristic	Maximum voltage	Current rating (amperes)	Cold resistance (ohms) 1/		Voltage drop (volts) 2/		Lead diameter	
				Minimum	Maximum	Minimum	Maximum	Inches ±(0.002)	mm ±(0.05)
FM11	A	125V	1/8A	1.89	2.31	.85	1.15	.025	.64
FM11	A	125V	1/4A	.639	.781	.59	.800	.025	.64
FM11	A	125V	3/8A	.378	.462	.527	.713	.025	.64
FM11	A	125V	1/2A	.252	.308	.488	.660	.025	.64
FM11	A	125V	3/4A	.153	.187	.145	.197	.025	.64
FM11	A	125V	1A	.112	.138	.157	.213	.025	.64
FM11	A	125V	1-1/2A	.072	.088	.153	.207	.025	.64
FM11	A	125V	2A	.0495	.0605	.144	.196	.025	.64
FM11	A	125V	2-1/2A	.0378	.0462	.125	.169	.025	.64
FM11	A	125V	3A	.0315	.0388	.139	.187	.025	.64
FM11	A	125V	4A	.0207	.0253	.110	.150	.025	.64
FM11	A	125V	5A	.0126	.0154	.087	.118	.025	.64
FM11	A	125V	7A	.009	.011	.087	.118	.025	.64
FM11	A	125V	10A	.0059	.007	.085	.110	.025	.64
FM11	A	50V	15A	.0036	.0044	.065	.087	.032	.82

1/ Cold resistance is measured at 10 percent or less of rated current at an ambient temperature of 25°C ±5°C.
2/ Voltage drop is measured at rated current after the fuse has been subjected to rated current for not less than 5 and not more than 10 minutes. The accuracy of the ammeters and voltmeters used shall be 1 percent or better.

Marking: Fuses shall be marked with the manufacturer's name or trademark and current rating only. A dot shall precede the manufacturer's name or trademark to distinguish the FM11 from the FM08.

QUALIFICATION

Qualification shall be in accordance with MIL-F-23419, table II specified herein, and any one of the three methods described below.

a. Sources not currently qualified to FM08 or FM04 must complete qualification using the column A of table II.

b. Sources currently qualified and using the same design and construction to style FM08 or FM04 must complete only those tests applicable to FM11 that have not been performed by previous qualification to FM08 or FM04.

c. Sources currently qualified to FM08 or FM04 but are using a different design and construction for FM11 must complete column A of table II.

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TABLE II. Qualification inspection.

Inspection	Number of sample units per current rating	FM11	FM08	FM04
		A	B	C
Group I				
Visual and mechanical examination	All	X	X	X
Continuity		X	X	X
Resistance		X	X	X
Voltage drop		X	X	
Current-carrying capacity		X	X	X
Group II	16 total			
Terminal strength	All	X	X	X
Overload interrupt		X		
200 percent at -55°C	4			
200 percent at +125°C	4			
300 percent at -55°C	4			
300 percent at +125°C	4			
Insulation resistance	All	X	X	X
Solderability	6	X	X	
Group III 1/	4			
Short circuit		X	X	X
Insulation resistance		X	X	X
Group IV	4			
Vibration, high frequency		X	X	X
Vibration, random		X		
Continuity		X	X	X
Shock		X	X	X
Continuity		X	X	X
Group V	4			
Salt spray (corrosion)		X	X	X
Overload interrupt (at room ambient temperature and maximum voltage rating)		X	X	X
Insulation resistance		X	X	X
Group VI	4			
Outgassing 2/		X		
Moisture resistance		X	X	X
Thermal shock		X	X	X
Resistance to soldering heat		X	X	
Current carrying capacity (at room ambient temperature)		X	X	X
Overload interrupt (at room ambient temperature and maximum voltage rating)		X	X	X
Insulation resistance		X	X	X

1/ The number of group III samples for qualification testing shall consist of four sample fuses of each of the current ratings as shown in table III. Short circuit interrupt test shall be conducted at each of the voltages indicated, and the fuses shall be capable of withstanding minimum short circuit as shown in table III. Insulation resistance test shall be as specified in MIL-F-23419.

2/ One sample representative of all current ratings.

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TABLE III. Short circuit tests.

Fuse type	Current rating	Short circuit current (dc)	DC voltage	Short circuit current (ac)	AC voltage (power factor 1.0)
FM11	10A	300A	125	50A	125
FM11	15A	300A	50	50A	32

EXTENT OF QUALIFICATION

a. Sources not currently qualified to FM08 or FM04 may receive qualification to FM08 and FM04 after completion of column A of table II and using the same design and construction for all three slash sheets.

b. Sources currently qualified to FM08 or FM04 may receive qualification to FM11 after completion of those test listed in column A which are not applicable to FM08 and FM04 and using the same design and construction for all three slash sheets

Quality conformance inspection: The following tests shall be conducted on 100 percent of the units, in the order shown, and shall replace the group A tests referenced in MIL-F-23419. Nonconforming units shall be removed from the lot.

a. Visual and mechanical. 1/

b. Resistance. 2/

c. Voltage drop. 2/

d. Thermal shock. 3/

e. Voltage drop. 4/

f. Seal. 5/

Revision letters are not used to denote changes due to the extensiveness of the changes.

1/ Inspection shall be by 10X magnification in accordance with MIL-F-23419.
2/ See table I.
3/ Testing temperature shall be from -55°C to +125°C. Fuses shall be monitored during cycling. Fuses that show evidence of physical damage or discontinuity in resistance greater than 10 microseconds in duration shall be removed from the lot. Fuses may be connected in series to monitor discontinuities.
4/ Shall not differ by more than 5 percent from value measured in voltage test performed prior to thermal shock above.
5/ Immerse in a fluorocarbon liquid bath at 125°C for 1 minute. No bubbles shall be detected. (Reference method 112, test condition D of MIL-STD-202.)

CONCLUDING MATERIAL

Preparing activity
NASA - NA

Agent:
DLA - ES

(Project 5920-K186)

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